

BIM-Guard Stair Testing — Property Reference

Every stair, landing, and railing/handrail property BIM-Guard can extract or compute, for use when preparing or reviewing a test IFC model. Covers IfcStairFlight, IfcStair, IfcSlab (PredefinedType = LANDING), and IfcRailing / IfcHandRail.

Two extraction layers feed these properties. **Layer 1** (Table 1) is read directly from the IFC file — standard property sets/quantities plus custom Shared Parameters — and depends entirely on Revit export settings (see the IFC Export Setting page). **Layer 2** (Table 2) is computed by BIM-Guard's own mesh-based stair geometry engine at analysis time; it needs correctly tessellated geometry for real IfcStairFlight/IfcRailing/IfcSlab(LANDING) entities, but no Revit-side property mapping at all.

Table 1 — Standard IFC Properties & Custom Shared Parameters

IFC Class	BIM-Guard Property	IFC Source	Export Requirement	Notes
IfcStairFlight	RiserHeight	Pset_StairFlightCommon.RiserHeight	IFC Common Property Sets	Direct match
IfcStairFlight	TreadDepth	Pset_StairFlightCommon.TreadLength	IFC Common Property Sets	Name mismatch — rule aliases to TreadLength automatically
IfcStairFlight	HeadroomClearance	Pset_StairFlightCommon.RequiredHeadroom	IFC Common Property Sets	Name mismatch, aliased — superseded by engine-computed MinHeadroom (Table 2) for most checks
IfcStairFlight	NumberOfRiser / NumberOfTreads	Pset_StairFlightCommon	IFC Common Property Sets	Exported automatically; not yet consumed by a shipped rule
IfcStairFlight / IfcStair	NosingLength	Pset_StairFlightCommon.NosingLength	IFC Common Property Sets	Aliases: Nosing, NosingProjection, NosingDepth
IfcStairFlight / IfcStair	WaistThickness	No standard property	Custom Shared Parameter	Aliases: Waist, StairWaist — needs a Shared Param mapped via the user-defined Pset file
IfcStairFlight / IfcStair	WalkingLineOffset	No standard property	Custom Shared Parameter	Aliases: WalklineOffset, WalkLineOffset
IfcStair	HandicapAccessible	No standard property	Custom Shared Parameter	Aliases: Accessible, IsAccessible, AccessibleRoute
IfcStair	HasNonSkidSurface	No standard property	Custom Shared Parameter	Aliases: NonSkidSurface, SlipResistant, SlipResistance, AntiSlip
IfcStair	FireExit	No standard property	Custom Shared Parameter	Aliases: IsFireExit, EmergencyExit
IfcStairFlight	Width	No standard property	Custom Shared Parameter	See BIMGuard_UserDefinedPsets.txt (Pset_BIMGuardStairFlight)
IfcStairFlight	FlightHeight	No standard property	Custom Shared Parameter	See BIMGuard_UserDefinedPsets.txt
IfcStairFlight	WinderTurnAngle / IndividualWinderAngle / WinderSetSeparation	No standard property	Custom Shared Parameter	See BIMGuard_UserDefinedPsets.txt — winder-specific, no automatic geometric equivalent yet

IFC Class	BIM-Guard Property	IFC Source	Export Requirement	Notes
IfcRailing	Reference / IsExternal	Pset_RailingCommon	IFC Common Property Sets	Exported automatically; not yet consumed by a shipped rule
IfcRailing	PredefinedType	Direct IFC attribute	Always exported	HANDRAIL vs GUARDRAIL/BALUSTRADE/FENCE/unset — gates which railing checks apply (Table 2, Group F)
IfcRailing	Height / HandrailHeight	No standard property	Custom Shared Parameter	See BIMGuard_UserDefinedPsets.txt (Pset_BIMGuardRailing)
IfcSlab (landing)	FireRating	Pset_SlabCommon.FireRating	IFC Common Property Sets	Direct match
IfcSlab (landing)	MaxSlope	Pset_SlabCommon.PitchAngle	IFC Common Property Sets	Name mismatch — rule aliases to PitchAngle
IfcSlab (landing)	HeadroomClearance	No standard property	Custom Shared Parameter	Superseded by engine-computed MinHeadroom (Table 2) for most checks
IfcSlab (landing)	Width — DO NOT USE for clear width	Qto_SlabBaseQuantities.Width	Base Quantities	Landmine: this is slab THICKNESS on every IfcSlab, landings included — never clear walking width. Use LandingClearWidth (Table 2) instead

Table 2 — Engine-Derived Geometry Properties

BIM-Guard Property	Measures	Unit
Group A — Per-flight riser/tread geometry (worst value within THIS flight)		
MinRiserHeight / MaxRiserHeight	Smallest / largest riser height detected in the flight	mm
RiserHeightDifference	Spread between the smallest and largest riser in the flight	mm
MinTreadDepth / MaxTreadDepth	Smallest / largest tread going (nosing-to-nosing) in the flight	mm
TreadDepthDifference (alias GoingDifference)	Spread between the smallest and largest going in the flight	mm
MinClearStairWidth (alias MinClearWidth)	Narrowest lateral clearance sampled along the flight's run — catches a local pinch point, not just the overall footprint width	mm
OpenRiserDetected (alias OpenRiser)	Whether any tread-to-tread transition lacks a closed riser face	boolean
TotalFlightRise / TotalFlightRun	Overall vertical rise / horizontal run of the whole flight	mm
FlightPitch	Overall flight angle from rise/run	deg
FlightSlopedLength	Straight-line sloped length of the flight (hypotenuse of rise/run)	mm
NumberOfTreadsDetected / NumberOfRisersDetected	Count of tread-top bands / mid-flight risers actually detected in the mesh	count
FlightStartElevation / FlightEndElevation	Absolute Z elevation at the bottom / top of the flight	mm
RiserHeights / TreadDepths	Full per-step lists (every riser / every going), not just min-max-difference — for exists/document checks, not numeric thresholds	list of mm
StepFormulaMin / StepFormulaMax	Smallest / largest per-step riser+going stride value ($2 \times \text{riser} + \text{going}$), paired per transition — not mixed from the flight's separate riser/going extremes	mm
StepFormulaValues	Full per-step $2 \times \text{riser} + \text{going}$ list, paired with that same transition's own riser/going	list of mm

BIM-Guard Property	Measures	Unit
Group B — Headroom (whole-model overhead search)		
MinHeadroom	Worst vertical clearance found above the flight's walking line or the landing's centroid, searched against every nearby slab/beam/flight/covering/roof in the model	mm
MinHeadroomLimitingGlobalId	GlobalId of the element causing that worst clearance	GUID

BIM-Guard Property	Measures	Unit
Group C — Whole-stairway uniformity (every flight of the same IfcStair pooled)		
StairRiserHeightDifference	Riser spread across ALL flights of this stairway, not just one flight	mm
StairTreadDepthDifference	Going spread across ALL flights of this stairway	mm
StairFlightCount	Number of flights pooled into this stairway's uniformity figures	count

BIM-Guard Property	Measures	Unit
Group D — Cross-referencing (flight ↔ landing ↔ railing ↔ stair linkage)		
ParentStairGlobalId	GlobalId of the IfcStair this flight/landing/railing belongs to (resolved by decomposition or proximity matching)	GUID
LandingBelow / LandingAbove	GlobalId of the landing connecting below / above this flight	GUID
ConnectsFlightBelow / ConnectsFlightAbove	GlobalId of the flight a landing connects below / above it	GUID
LandingLevelMismatch	Worst elevation gap between a landing and the flight(s) it's matched to	mm
HandrailCountOnFlight / GuardCountOnFlight	Number of HANDRAIL-type / guard-type railings matched to this flight	count
HostElementGlobalId	GlobalId of the flight or landing a railing is matched to (nearest by plan proximity)	GUID

BIM-Guard Property	Measures	Unit
Group E — Landing geometry (IfcSlab, PredefinedType = LANDING)		
LandingClearWidth / LandingClearLength	Min-rotated-rectangle clear walking dimensions of the landing's own footprint mesh (NOT Qto_SlabBaseQuantities.Width — see Table 1 landmine)	mm
LandingClearArea	Landing footprint area	mm ²
LandingElevation	Absolute Z elevation of the landing's top surface	mm
LandingSlope	Tilt of the landing's own walking surface (isolated top faces only, not the whole slab's bounding box — avoids misreading slab thickness as slope)	deg

BIM-Guard Property	Measures	Unit
Group F — Handrail / guard geometry (IfcRailing / IfcHandRail)		
HandrailMinHeight / HandrailMaxHeight	Lowest / highest top-of-rail elevation relative to the walking surface, sampled along the rail's run	mm
HandrailHeightVariation	Spread between the highest and lowest top elevation sampled	mm
HandrailPathLength	Straight-line run-axis length of the rail (undercounts a curved rail — flagged in warnings when curvature is detected)	mm
HandrailContinuousSegments	Number of physically continuous material segments detected along the run	count
HandrailMaxGapLength	Length of the largest real break between continuous segments	mm
HandrailGapLocations	Run-axis (start, end) position of every detected gap	list of mm pairs
HandrailMinBottomElevation	Lowest bottom-of-rail elevation sampled	mm
BottomClearGap	Median clear gap between the rail's bottom member and the walking surface (median, not minimum — avoids floor-anchored posts masking a real elevated gap)	mm
HandrailProfileLateral / HandrailProfileVertical	Coarse cross-section width / height of the rail profile near mid-run	mm
MaxOpening	Guard-type railings only: largest horizontal infill/baluster gap found across several sampled heights	mm
GuardMaxOpening	Guard-type railings only: the worse of MaxOpening and BottomClearGap, for a single “worst opening anywhere” check	mm
HandrailExtensionBottom / HandrailExtensionTop	How far this rail's own path reaches past its host flight's bottom / top tread nosing, projected onto the flight's own walking direction. Positive extends past that end; negative falls short of reaching it (a real, worse condition, not clipped to zero)	mm

Alias Resolution (already handled automatically — no rule or Revit change needed)

Rule Property Name	Also Tries (resolved automatically)	IFC Class
NosingLength	Nosing, NosingProjection, NosingDepth	IfcStairFlight, IfcStair
WaistThickness	Waist, StairWaist	IfcStairFlight, IfcStair
HandicapAccessible	Accessible, IsAccessible, AccessibleRoute	IfcStair
HasNonSkidSurface	NonSkidSurface, SlipResistant, SlipResistance, AntiSlip	IfcStair
Headroom	RequiredHeadroom, HeadroomClearance, ClearHeight, ClearanceHeight	IfcStair
WalkingLineOffset	WalklineOffset, WalkLineOffset	IfcStairFlight, IfcStair
FireExit	IsFireExit, EmergencyExit	IfcStair

Known v1 Limitations

- Winder / curved flights: overall turning angle and rise/run still resolve, but per-winder tread depth at the inner / walking-line / outer edges is not yet computed. A lateral tread-centroid drift beyond ~75mm is detected internally and flags the flight's warnings as “winder suspected”, but this flag is not yet exposed as a queryable rule property.
- Guards (MaxOpening / GuardMaxOpening): computed as the largest horizontal gap at a sampled height, not a true multi-directional sphere-passing simulation. Baluster centre-to-centre spacing and the triangular stair-nosing opening are not yet computed as their own figures.
- Handrail/guard path length (HandrailPathLength) is a straight-line run-axis approximation; a curved rail's true swept length is undercounted (flagged in warnings when curvature is detected).
- Headroom (MinHeadroom) is sampled at discrete points — one per detected tread nosing for a flight, one at the plan centroid for a landing — not scanned continuously across the full walking surface. An obstruction that misses every sample point is not caught.
- Whole-stairway uniformity (Group C) requires flights to actually resolve to the same stairway, either via a real IfcStair decomposition relationship or proximity-based fallback grouping — flights that don't group correctly report uniformity figures for themselves alone.

Computed Internally but Not Yet Queryable by Name

- Curvature drift distance and the “winder suspected” flag (surfaced only in the element's warnings text)
- Per-height opening gap series behind MaxOpening (only the worst value is exposed as a property)
- The exact XYZ location of the worst headroom clearance point (only the limiting element's GlobalId is exposed)
- Per-tread nosing world coordinates, per-tread elevations, and open-riser run positions (used internally for headroom sampling and warning text)
- Each element's own world bounding box (used internally for cross-referencing; not itself a rule property)

See also: [frontend/public/downloads/IFC_Export_Setting.json](#), [BIMGuard_UserDefinedPsets.txt](#), and [README.md](#) for how to configure a Revit export that populates Table 1's properties, and [docs/ifc-property-mapping.md](#) for the full cross-domain property mapping reference.